

Corentin Le Guen

Montreal QC, Canada | +1 438 530 4307 | leguen.corentin@protonmail.com |
<https://www.linkedin.com/in/corentinleguen/>

Senior Backend Developer & Data Engineer

Software developer with experience in the banking industry, specialized in creating robust and scalable solutions for complex business needs. Passionate about leveraging technology to solve challenging problems and create value. Expertise in back-end development, real-time data processing, cloud deployment, and data analytics. Modernized architectures, developed critical applications, and implemented cutting-edge tools to support high-frequency trading and financial analysis. Combines technical proficiency with a strategic mindset to drive performance and innovation. Thrives in collaborative environments, delivering projects and mentoring others.

Technologies

Back-end: Python, Java, Kotlin, Q/KDB+

Front-end: Vue3, TailwindCSS, Flowbite

Databases: Cassandra, SQLite3, MySQL, PostgreSQL, CockroachDB, MongoDB, Redis

Testing: Gatling, Mockito, Junit, Cucumber, Sonar

Tools & Frameworks: Flask, FastAPI, Splunk, DataDog, Git, Maven, GitLab / GitHub / Bitbucket, Postman, PyCharm, IntelliJ, Docker, Jenkins

Professional Experience

Alithya

Since June 2025

Senior Backend Developer & Data Engineer

Client: Desjardins. Played a key role in designing and implementing a secure, scalable architecture and backend services for banking file processing.

- Designed a resilient architecture for secure banking file processing and developed a proof of concept to validate the solution.
- Built Kotlin-based microservices and RabbitMQ pipelines for asynchronous file processing and automated email delivery.
- Participated in code reviews, including reviewing and approving pull requests.
- Contributed to the refinement of technical requirements.

Stack: Kotlin | MSSQL | RabbitMQ | Angular | GitHub | Docker | Azure

Morgan Stanley

March 2023 - March 2025

Data Scientist, Strategist & Data Analyst

Developed and maintained data tools and applications to support trading operations, focusing on real-time insights and reliability.

- Designed and maintained dashboards for trading desks, delivering actionable performance metrics and analytics to support strategic decision-making.
- Partnered with cross-functional teams to gather requirements and design scalable, data-driven solutions.

- Leveraged Q/KDB+ for real-time data processing and analytics, enabling high-frequency trading strategies and high-volumes API insights.
- Performed in-depth data audits and quality assessments to ensure data integrity and reliability across systems.
- Delivered actionable insights and data-driven recommendations to optimize processes, improve reliability, and mitigate operational risks.

Stack: Q/KDB+ | Python | BitBucket | TeamCity | Docker | ClikView

FX Innovation

August 2021 - March 2023

Senior Java Developer & Tech Lead

Client: Desjardins. Tech lead on the modernization team. Java/Spring API for Oracle security. Deployment on Weblogic and Microsoft Azure. CI/CD on BitBucket / GitHub and Concourse. Testing with JUnit, Cucumber, Gherkin and Sonar. Project management with Jira and Confluence for documentation.

- Participated in modernizing the architecture.
- Updated and maintained an app to process files with Java/Spring.

Stack: Java | Spring | Swagger | Git | BitBucket | GitHub | Sonar | Junit | Cucumber | Gherkin | Concourse | Postman | Jira | Azure | Docker

Systematix

September 2019 - August 2021

Senior Fullstack Developer

Client: National Bank of Canada. Maintained apps with NodeJS. Deployment with Ansible and Terraform. Tests with Axios, Mocha and Sonar. App monitoring with Splunk and DataDog. Installation and OpenText Exstream for document generation. CI/CD on BitBucket, Jenkins, Terraform on OpenShift.

- Updated Dockerfiles. Updated deployment scripts with VI on Linux servers.

Stack: NodeJS | FeathersJS | Express | Axios | Mocha | OpenShift | Jenkins | BitBucket | Ansible | Nexus | Vault | Jira | Confluence | Bash | Linux | VI | Docker | Dockerfile | Sonar | Splunk | DataDog | PostgreSQL | Postman | Docker | OpenText Exstream | Control-M | MFT

Full stack Developer

Client: McGill University. Maintained a RabbitMQ app with Java and Junit. Added a security layer on API Connect.

- Created an algorithm and an app to confront JSON files with JQuery. Deployed on OpenShift and GitLab.

Stack: Java | RabbitMQ | Junit | JQuery | API Connect | NodeJS | Jira | Confluence | Docker

Ippon Technologies, France

November 2018 - August 2019

Full stack Developer | 10 months

Client: BNP Paribas. Maintained a customer account consultation app on agile/scrum process. Testing with JUnit and Cucumber. Testing load on servers with Gatling. Helped two Java junior developer.

- Developed a Java, Spring app. Created producers and consumers Kafka. Upgraded Cassandra data models.

Stack: Java | Spring | JUnit | Cucumber | Gatling | Kafka | Cassandra | Solr | Git | Jira | Bash | Linux servers

CGI, France

September 2018 - November 2018

Client: SFR (Phone operator)

Full stack Developer

Migrate ERP (Enterprise Resource Planning). Migrate Oracle servers. Maintained an app of professional accounts management. Upgraded a Java app (upgrade Java version from Java 6 to Java 8).

Stack: Java | Oracle | Git | Bash

Altitude Infrastructure, France

July 2017 - August 2018

Developer

Application manager for phone operators. Maintained a C# app and developed a new C# app with .NET. Created a Front-end app to visualize KPI with C# and Bootstrap. Installed a new ERP (Enterprise Resource Planning) with transcoding tables. Migrated ERP from Adonix to SageX3. User testing.

Stack: C# | ASP.NET | Bootstrap 3 | JQuery | HTML | CSS | Bash | Git | Redmine | SQLServer

Saagie, France

October 2016 - June 2017

Developer | Intern

Logging app to track customers journey on the website with Java, Hadoop, Cloudera and Hive. Logging javascript library with JQuery. UX/UI improvement.

Stack: Java | Kafka | Hive | Hadoop | Cloudera | Bash | Javascript | JQuery

Personal Projects

Rose

An AWS S3 Object Store space tracker using Rust, CockroachDB, Docker and Amazon S3 2025
github.com/CorentinLeGuen/rose

Personnal Website and portfolio

2025

A personnal blog / website using NodeJS, Sveltekit, TailwindCSS, Netlify and Flowbite
<https://cleguen.com>

Project "Life Gamera"

2023

The Conway's Game of Life implementation using Raylib and Python.
<https://github.com/CorentinLeGuen/lifegamera>

Project "Lotto 6.49 API"

2022

An API containing Lotto draws since 1982-06-12. Python3, Flask, SQLite, Docker
More here: <https://github.com/CorentinLeGuen/lotto-6-49-api>

Project "Python on Page"

2022

Running Python on browser using WebAssembly. Python3, Flask, WebAssembly

<https://github.com/CorentinLeGuen/python-on-page>

Project “pycasso”

2022

A project to transform a picture into ASCII code using Python and Numpy.

<https://github.com/CorentinLeGuen/pycasso>

Renew project “Le Compte est Bon”

2022

Renew the project from 2013 with new technologies to solve the algorithm using Python

More here: <https://github.com/CorentinLeGuen/le-compte-est-bon>

Project “Scrap and analyse”

2021 & 2022

A project to scrap data from websites and analyse them to predict next incoming update. Project hosted on a VPS (Virtual Proxy Server) on Linux. Python3, Pandas, BeautifulSoup4, Flask, Bash

Projetct “Loto Quebec Last Result API”

2021

A project to scrap Data from Loto 6.49 website using Java

<https://github.com/CorentinLeGuen/LotoQuebec649LastResults>

Project “array list” in Go

2020

A project made in Go to provide a Java arraylist implementation like. This project have been cloned more than 250 times and has 1 fork. Go. Code source here:

<https://github.com/CorentinLeGuen/le-compte-est-bon>

Project “Simple KeyLogger”

2020

A small project to save input keys into a log file. Python3

<https://github.com/CorentinLeGuen/SimpleKeyLogger>

Project “Where is ISS?”

2018

A front-end project to track the ISS position on earth globe. Javascript, JQuery, PlanetaryJS

Technology used: <https://miniature.earth/demo/space-view.htm>

Project “Le Compte est Bon”

2013

Created an algorithm to solve the game and a GUI with GTK. Made with Python2 and PyGTK.

More here: https://fr.wikipedia.org/wiki/Des_chiffres_et_des_lettres#Le_Compte_est_Bon

Education and Professional Development

GitHub Action – certification, Microsoft

Q4 2025

AWS Certified Developer – Associate, Amazon Webservices certificate

2025

Learning AI Through Visualization, Columbia University

2025

Data Science & Machine Learning with Python, Udemy Online course at udemy.com

2022

Certificate of Language Training in English, ACA Online course at lesateliers.ca

2021

Specialization: Programming with Google Go, Coursera Online course at coursera.org

2019

Intermediate R Course, DataCamp Online course at datacamp.com

2019

Deep Learning, National Conservatory of “Arts et Métiers”, France Online course

2018

Master degree of Computer Software Engineering, University of Rouen Normandie, France	2018
Licence Diploma of Development, University of Rouen Normandie, France	2016
Introduction to C++ Development, Polytechnic School of Lausanne, Swiss Online course at coursera.org	2013
Introduction to Java Development, Polytechnic School of Lausanne, Swiss Online course at coursera.org	2013

Languages: Bilingual in English and French